SKM195GAL126D



SEMITRANS® 2

Trench IGBT Modules

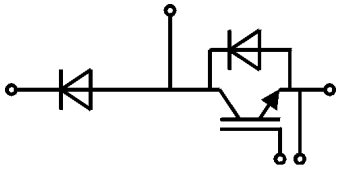
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Features

- Trench = Trenchgate technology
- $V_{CE(sat)}$ with positive temperature coefficient
- High short circuit capability, self limiting to 6 x I_C
- UL recognized, file no. E63532

Typical Applications*

- Electronic welders
- DC/DC – converter
- Brake chopper
- Switched reluctance motor



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Absolute Maximum Ratings

Symbol	Conditions		Values	Unit
IGBT				
V_{CES}	T_j = 25 °C		1200	V
I_C	T_j = 150 °C	T_c = 25 °C	220	A
		T_c = 80 °C	164	A
I_{Cnom}			150	A
I_{CRM}	I_{CRM} = 2xI_{Cnom}		300	A
V_{GES}			-20 ... 20	V
t_{psc}	V_{CC} = 600 V V_{GE} ≤ 15 V V_{CES} ≤ 1200 V	T_j = 125 °C	10	µs
T_j			-40 ... 150	°C
Inverse diode				
I_F	T_j = 150 °C	T_c = 25 °C	143	A
		T_c = 80 °C	98	A
I_{Fnom}			100	A
I_{FRM}	I_{FRM} = 2xI_{Fnom}		200	A
I_{FSM}	t_p = 10 ms, sin 180°, T_j = 25 °C		1110	A
T_j			-40 ... 150	°C
Freewheeling diode				
I_F	T_j = 150 °C	T_c = 25 °C	143	A
		T_c = 80 °C	98	A
I_{Fnom}			100	A
I_{FRM}	I_{FRM} = 2xI_{Fnom}		200	A
I_{FSM}	t_p = 10 ms, sin 180°, T_j = 25 °C		1110	A
T_j			-40 ... 150	°C
Module				
$I_{t(RMS)}$			200	A
T_{stg}			-40 ... 125	°C
V_{isol}	AC sinus 50 Hz, t = 1 min		4000	V

Characteristics

Symbol	Conditions		min.	typ.	max.	Unit
IGBT						
$V_{CE(sat)}$	I_C = 150 A V_{GE} = 15 V chiplevel	T_j = 25 °C		1.71	2.10	V
		T_j = 125 °C		2.00	2.45	V
V_{CE0}	chiplevel	T_j = 25 °C		1	1.2	V
		T_j = 125 °C		0.9	1.1	V
r_{CE}	V_{GE} = 15 V chiplevel	T_j = 25 °C		4.7	6	mΩ
		T_j = 125 °C		7.3	9	mΩ
$V_{GE(th)}$	V_{GE}=V_{CE}, I_C = 6 mA		5	5.8	6.5	V
I_{CES}	V_{GE} = 0 V V_{CE} = 1200 V	T_j = 25 °C			2	mA
		T_j = 125 °C				mA
C_{ies}	V_{CE} = 25 V V_{GE} = 0 V	f = 1 MHz		10.8		nF
C_{oes}		f = 1 MHz		0.56		nF
C_{res}		f = 1 MHz		0.49		nF
Q_G	V_{GE} = - 8 V...+ 20 V			1380		nC
R_{Gint}	T_j = 25 °C			5		Ω

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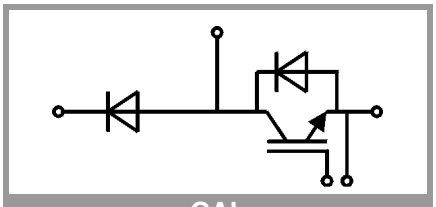
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Features

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Typical Applications*

- Electronic welders
- DC/DC – converter
- Brake chopper
- Switched reluctance motor



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Characteristics						
Symbol	**Conditions**		**min.**	**typ.**	**max.**	**Unit**
$t_{d(on)}$	V_{CC} = 600 V I_C = 150 A V_{GE} = +15/-15 V $R_{G\ on}$ = 2 Ω $R_{G\ off}$ = 2 Ω	T_j = 125 °C		280		ns
t_r		T_j = 125 °C		50		ns
E_{on}		T_j = 125 °C		16		mJ
$t_{d(off)}$		T_j = 125 °C		560		ns
t_f		T_j = 125 °C		70		ns
E_{off}		T_j = 125 °C		24.5		mJ
$R_{th(j-c)}$	per IGBT				0.16	K/W
Inverse diode						
$V_F = V_{EC}$	I_F = 100 A V_{GE} = 0 V chiplevel	T_j = 25 °C		2.00	2.50	V
		T_j = 125 °C		1.80	2.30	V
V_{F0}	chiplevel	T_j = 25 °C		1.1	1.45	V
		T_j = 125 °C		0.85	1.2	V
r_F	chiplevel	T_j = 25 °C		9	11	mΩ
		T_j = 125 °C		9.5	11	mΩ
I_{RRM}	I_F = 100 A di/dt_{off} = 2200 A/µs V_{GE} = -15 V V_{CC} = 600 V	T_j = 125 °C		86		A
Q_{rr}		T_j = 125 °C		17		µC
E_{rr}		T_j = 125 °C		5.8		mJ
$R_{th(j-c)}$	per diode				0.32	K/W
Freewheeling diode						
$V_F = V_{EC}$	I_F = 100 A V_{GE} = 0 V chiplevel	T_j = 25 °C		2.00	2.50	V
		T_j = 125 °C		1.80	2.30	V
V_{F0}	chiplevel	T_j = 25 °C		1.1	1.45	V
		T_j = 125 °C		0.85	1.2	V
r_F	chiplevel	T_j = 25 °C		9	11	mΩ
		T_j = 125 °C		9.5	11	mΩ
I_{RRM}	I_F = 100 A di/dt_{off} = 2200 A/µs V_{GE} = ±15 V V_{CC} = 600 V	T_j = 125 °C		86		A
Q_{rr}		T_j = 125 °C		17		µC
E_{rr}		T_j = 125 °C		5.8		mJ
$R_{th(j-c)}$	per Diode				0.32	K/W
Module						
L_{CE}				30		nH
$R_{CC'+EE'}$	terminal-chip	T_C = 25 °C		0.75		mΩ
		T_C = 125 °C		1		mΩ
$R_{th(c-s)}$	per module			0.04	0.05	K/W
M_s	to heat sink M6		3		5	Nm
M_t		to terminals M5	2.5		5	Nm
						Nm
w					160	g

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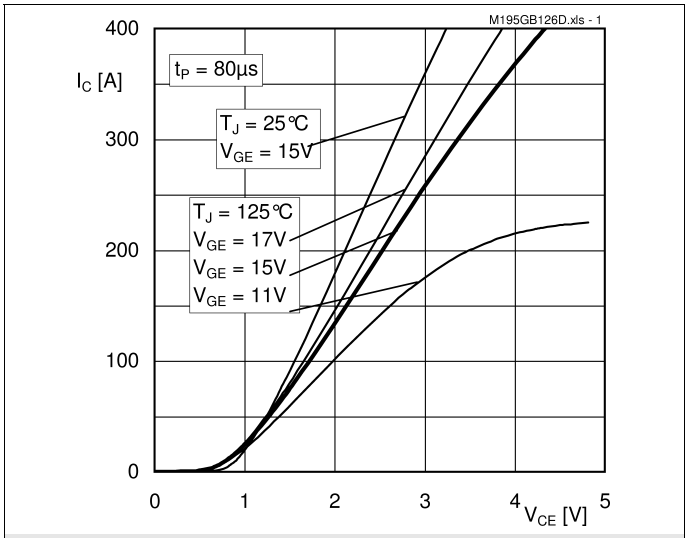


Fig. 1: Typ. output characteristic, inclusive $R_{CC'+EE'}$

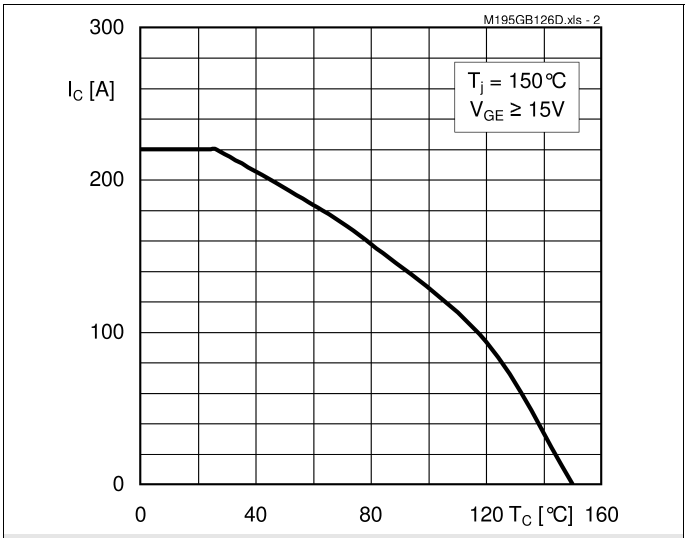


Fig. 2: Rated current vs. temperature $I_C = f(T_C)$

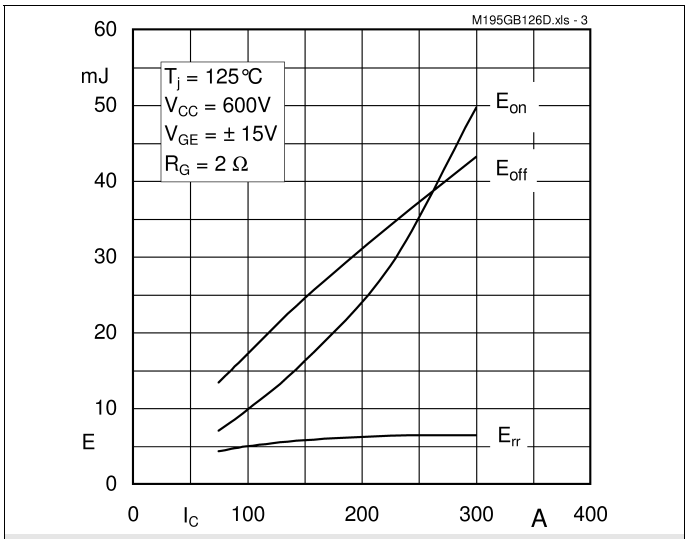


Fig. 3: Typ. turn-on /-off energy = f (I_C)

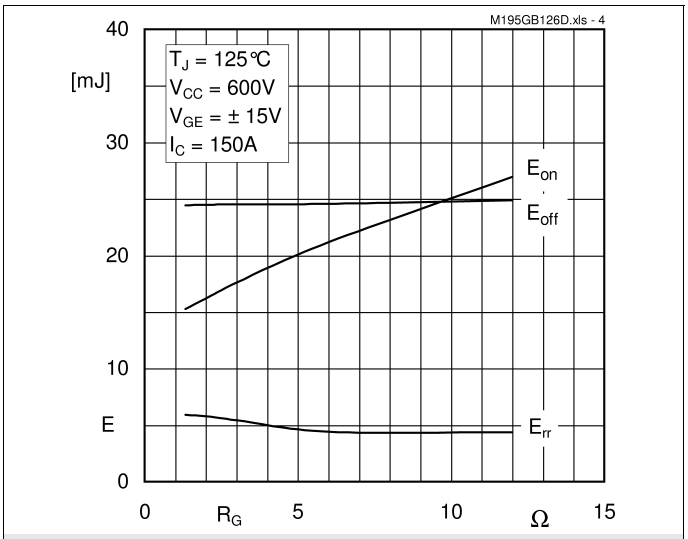


Fig. 4: Typ. turn-on /-off energy = f (R_G)

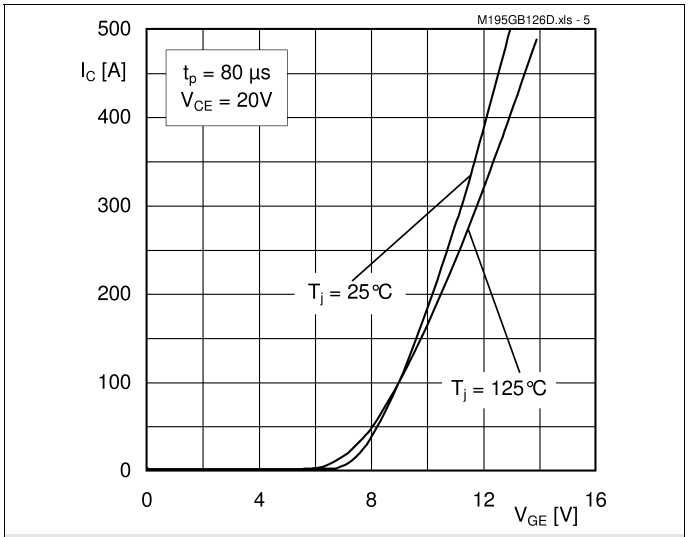


Fig. 5: Typ. transfer characteristic

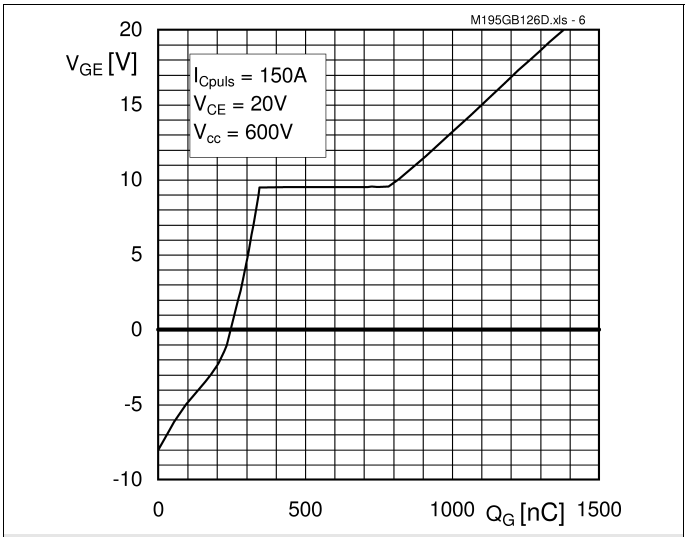


Fig. 6: Typ. gate charge characteristic

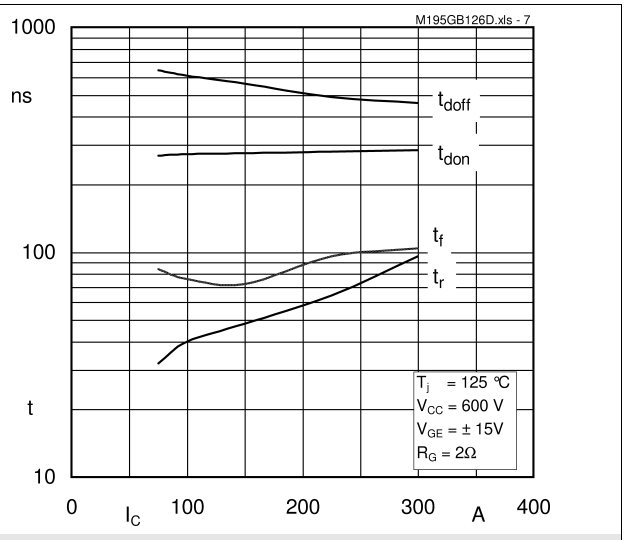


Fig. 7: Typ. switching times vs. I_C

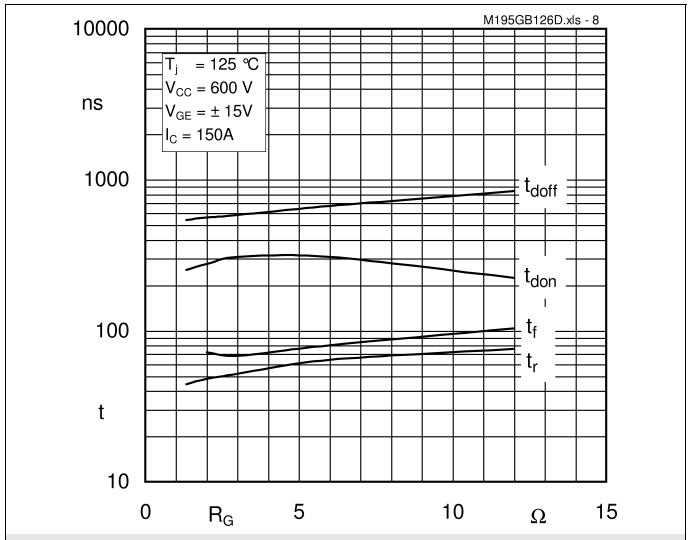


Fig. 8: Typ. switching times vs. gate resistor R_G

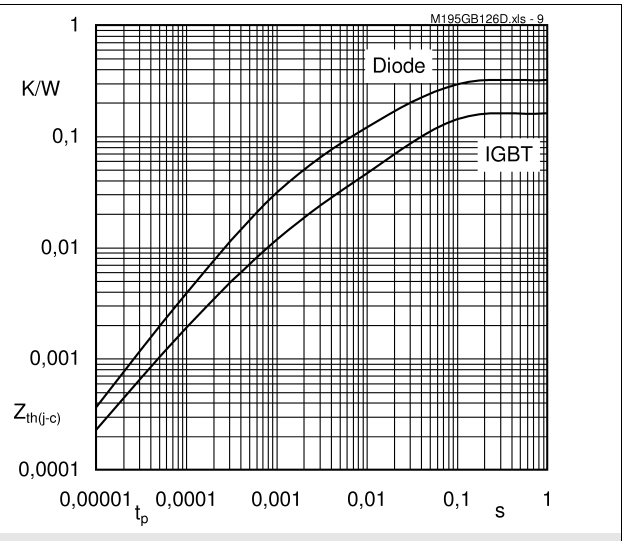


Fig. 9: Transient thermal impedance

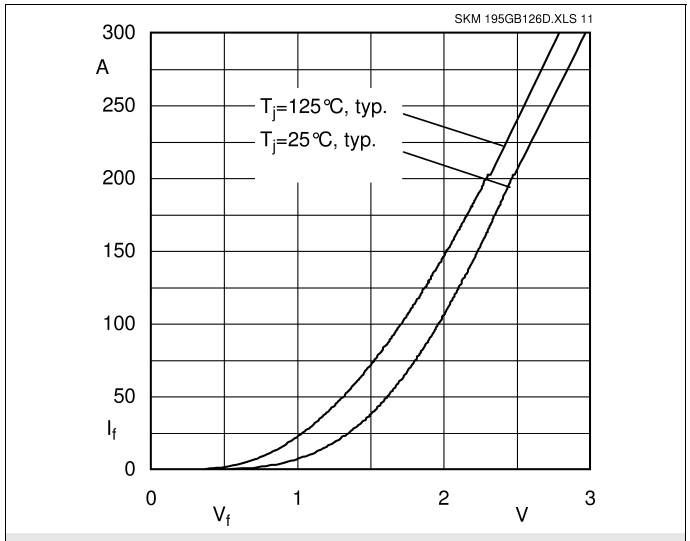


Fig. 10: Typ. CAL diode forward charact., incl. $R_{CC'+EE'}$

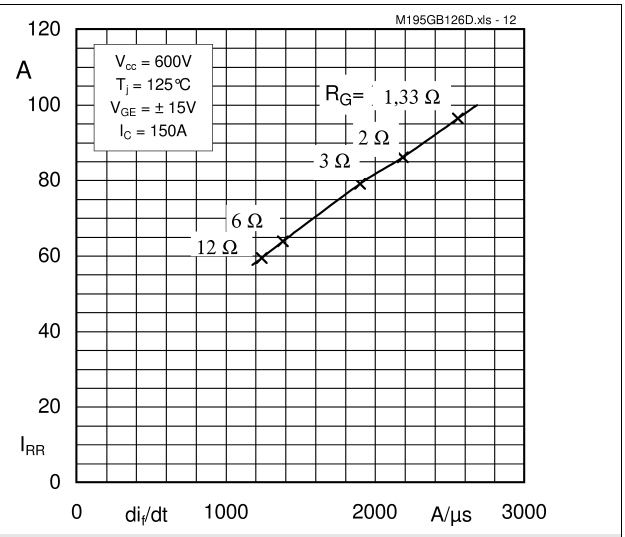


Fig. 11: Typ. CAL diode peak reverse recovery current

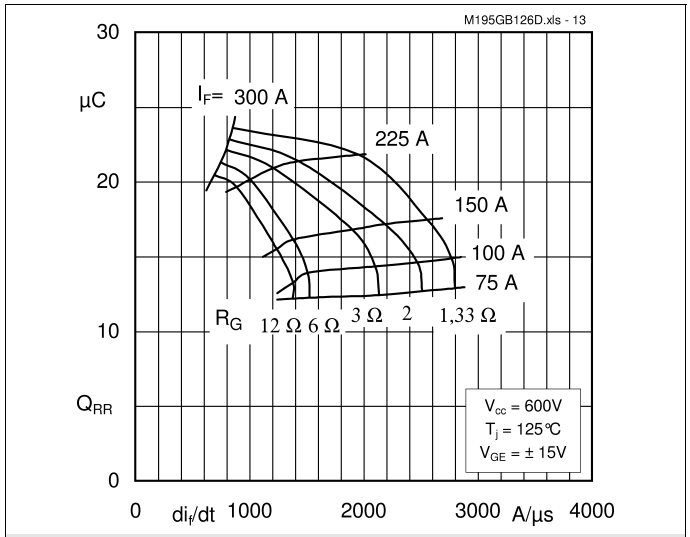
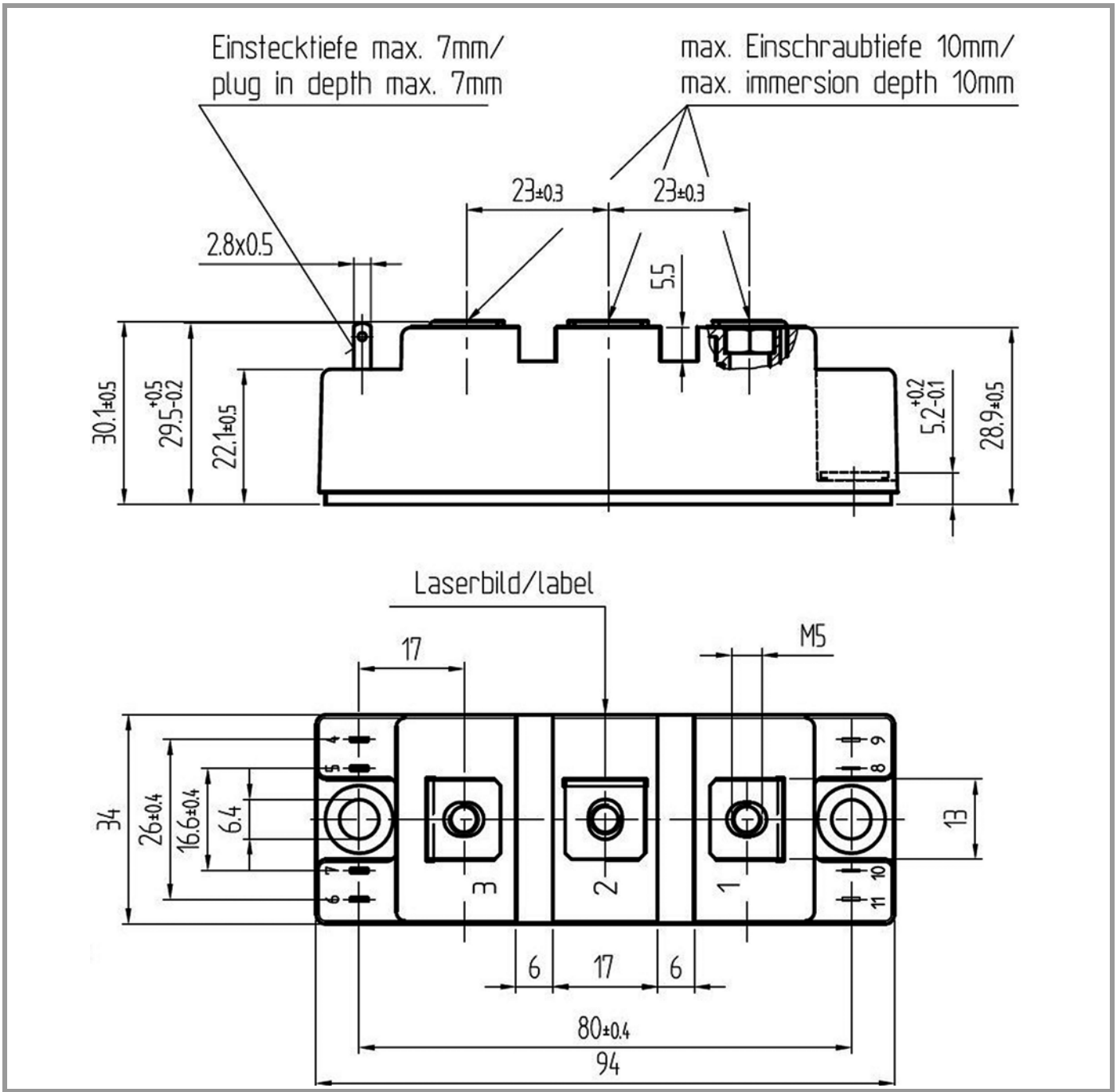
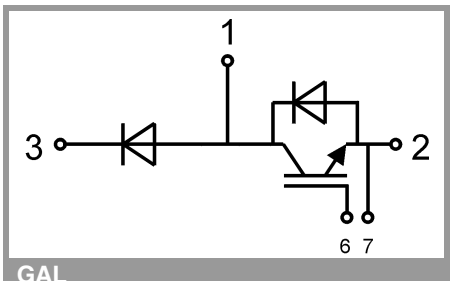


Fig. 12: Typ. CAL diode peak reverse recovery charge

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This is an electrostatic discharge sensitive device (ESDS), international standard IEC 60747-1, Chapter IX

* The specifications of our components may not be considered as an assurance of component characteristics. Components have to be tested for the respective application. Adjustments may be necessary. The use of SEMIKRON products in life support appliances and systems is subject to prior specification and written approval by SEMIKRON. We therefore strongly recommend prior consultation of our staff.